

Technology Stack

Date	15 JULY 2026
Project Name	Smart Lender: AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack

The Smart Lender application is developed using Python and Flask for the backend, with HTML, CSS, Bootstrap, and JavaScript for the frontend to create a responsive web interface. Machine learning models are built using Scikit-learn and XGBoost, while Pandas and NumPy are used for data preprocessing. The trained XGBoost model is saved using Pickle for real-time loan prediction. The application uses a CSV dataset for training and is deployed on IBM Cloud. Git and GitHub are used for version control, making the project easy to manage and maintain.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, Bootstrap, JavaScript	Used to build a responsive and user-friendly web interface where users can enter loan applicant details and instantly view loan approval predictions.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask provides a lightweight web framework for integrating the trained machine learning model with the web application and handling prediction requests efficiently</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset (Pandas DataFrame)</i>	<i>Applicant data is stored and processed using CSV files during model training. Pandas is used for data preprocessing and manipulation.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Cloud</i>	<i>IBM Cloud offers reliable and scalable cloud hosting for deploying the Flask application and making the prediction service accessible online.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git and GitHub enable version control, collaborative development, project tracking, and easy deployment management.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn, Pickle</i>	<i>Scikit-learn and XGBoost are used for training and evaluating machine learning models. Pandas and NumPy handle data processing, Matplotlib and Seaborn are used for visualization, and Pickle is used to save the trained XGBoost model for deployment.</i>